

House Price Prediction Using Linear Regression on California Housing Dataset

1. Introduction

Objective of the Project

The objective of this project is to build and evaluate a Linear Regression model for predicting house prices using the California Housing Dataset. The project demonstrates the complete Machine Learning workflow, including data loading, exploratory data analysis (EDA), data preprocessing, model training, evaluation, and result interpretation.

Machine Learning models are widely used in the real estate industry to estimate property values based on various factors. In this project, Linear Regression is used as a supervised learning algorithm to understand the relationship between housing features and house prices.

Dataset Used

The California Housing Dataset is a popular dataset available in Scikit-learn. It contains information collected from the 1990 California census. The dataset consists of multiple housing-related attributes and a target variable representing the median house value.

The dataset contains the following features:

- MedInc – Median income in a block group
- HouseAge – Median house age
- AveRooms – Average number of rooms
- AveBedrms – Average number of bedrooms
- Population – Population of the block group
- AveOccup – Average occupancy
- Latitude – Latitude coordinate
- Longitude – Longitude coordinate

Target Variable:

- HousePrice – Median house value

2. Exploratory Data Analysis (EDA)

Dataset Shape

The California Housing Dataset contains:

- 20,640 rows (samples)
- 9 columns (8 features + 1 target variable)

The dataset was checked for missing values, and no missing data was found.

Feature Description

Each feature provides useful information about housing characteristics and geographical location. Understanding these variables helps identify which factors influence house prices the most.

Correlation Heatmap

A correlation heatmap was generated to analyze relationships between features and the target variable. The heatmap revealed that:

- Median Income (MedInc) has the strongest positive correlation with house prices.
- Latitude and Longitude show noticeable relationships with housing values.
- Some features have weak correlations, indicating a limited direct effect on price prediction.

Distribution Plots

Distribution plots were created to understand the spread of housing prices and other variables.

Observations:

- House prices are not perfectly normally distributed.
- Most houses fall within a moderate price range.
- A few high-priced houses create a right-skewed distribution.

Scatter plots between Median Income and House Price further confirmed that higher-income areas generally have higher house values.

3. Model Building

Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio:

- Training Data: 80%
- Testing Data: 20%

The training dataset was used to train the model, while the testing dataset was used to evaluate its performance on unseen data.

Linear Regression Algorithm

Linear Regression is one of the simplest and most commonly used supervised Machine Learning algorithms. It predicts a continuous target variable by establishing a linear relationship between input features and the target variable.

The mathematical form of Linear Regression is:

$$\text{House Price} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

where:

- β_0 is the intercept
- β_1 to β_n are coefficients
- X_1 to X_n are input features

The model was trained using Scikit-learn's `LinearRegression()` class.

4. Results

The trained model was evaluated using standard regression metrics.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

$$\text{MAE} = 0.5332001304956557$$

Root Mean Squared Error (RMSE)

RMSE measures prediction error while penalizing larger errors more heavily.

$$\text{RMSE} = 0.7455813830127762$$

R² Score

R² Score indicates how well the model explains the variability of the target variable.

$$R^2 = 0.575787706032451$$

Interpretation

- The model achieves a moderate level of prediction accuracy.
- An R² score of approximately 0.57 indicates that around 57% of the variation in house prices is explained by the model.
- The model captures major trends in the dataset but still leaves room for improvement.

5. Conclusion

In this project, a Linear Regression model was successfully developed to predict house prices using the California Housing Dataset.

Key findings include:

- The complete Machine Learning workflow was implemented successfully.
- Exploratory Data Analysis helped understand the dataset and relationships among variables.
- Median Income was identified as one of the strongest predictors of house prices.
- The Linear Regression model produced reasonable predictions and demonstrated the usefulness of regression techniques for real-world problems.

Future Improvements

The model's performance can be further improved using advanced techniques such as:

- Random Forest Regressor
- XGBoost Regressor
- Feature Engineering
- Hyperparameter Tuning
- Cross Validation
- Outlier Detection and Removal

Overall, this project provided practical experience in data analysis, visualization, model training, evaluation, and deployment preparation using Python and Scikit-learn.